UNIT QUESTIONS

SKILLS CRITICAL THINKING 6

1 **a** Which of the following quantities is not a vector quantity?

A force

B area

C displacement

D velocity

(1)

b Which of the following quantities is not a scalar quantity?

A speed

B your age

C volume

D acceleration

(1)

c Which of the following statements about the motion of an object on which unbalanced forces act is false?

A The object could continue moving at constant speed in a straight line.

B The object could accelerate.

C The object could slow down.

D The object could change the direction in which it is moving.

(1)

d A ball is thrown vertically upwards in the air. Which of the following statements about the motion of the ball is false?

A The ball will be travelling as fast as it was when it gets back to the ground as it was when first thrown upwards.

B The ball will be stationary for an instant at the highest point in its flight.

C The direction of the force on the ball changes so the ball falls back to the ground.

D The direction of the acceleration is always downwards.

(1)

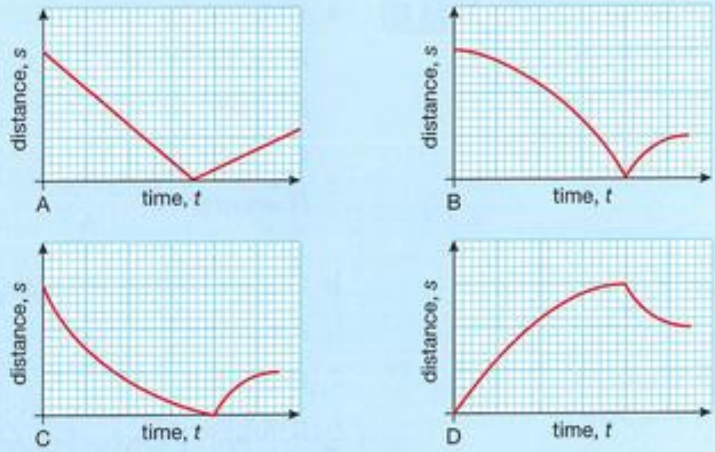
(Total for Question 1 = 4 marks)

SKILLS ANALYSIS

2 A tennis ball is dropped from a height of 3 m and bounces back to a height of 1 m after hitting the ground ($s = 0$).

6

a Which of the following distance–time graphs shows this motion correctly? **(1)**



SKILLS CRITICAL THINKING 7

b **i** Explain the difference between the terms distance and displacement. **(2)**

SKILLS PROBLEM SOLVING 8

ii What is the displacement of the tennis ball at the end of the described motion? **(1)**

9

c **i** Use the equation $v^2 = u^2 + 2as$ to calculate the velocity of the tennis ball when it hits the ground. **(4)**

7

ii What is the average speed of the tennis ball while it is falling to the ground? **(2)**

8

iii Calculate how long it takes for the tennis ball to reach the ground. **(2)**

SKILLS CRITICAL THINKING 6

d How can the distance travelled by the tennis ball be found from a velocity–time graph of its motion? **(1)**

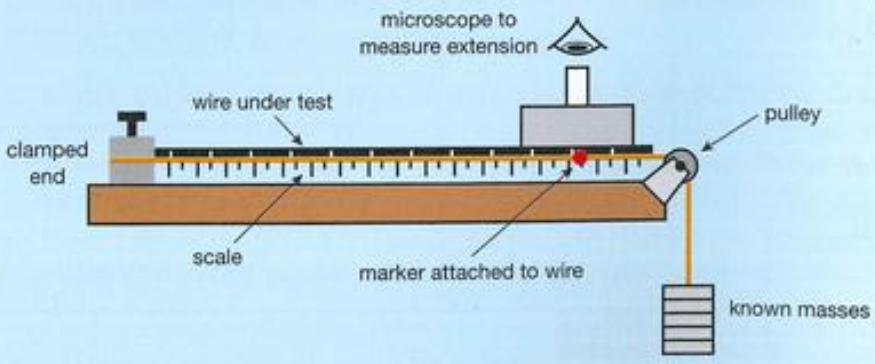
SKILLS INTERPRETATION 8

e Sketch a velocity–time graph for the tennis ball from the time that it is released to the time that it has rebounded to 1 m after the first bounce. Assume that the time that the ball is in contact with the ground is negligible (too short to show on your graph). Include as much numerical detail as you can without doing any further calculations. **(6)**

(Total for Question 2 = 19 marks)

SKILLS CRITICAL THINKING

3 A student uses the following piece of apparatus to investigate how the extension of a length of wire varies with the force applied to it.



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4

5

SKILLS DECISION MAKING

SKILLS EXECUTIVE FUNCTION 6

a i Name the independent variable in this investigation. **(1)**

ii Name the dependent variable in this investigation. **(1)**

iii Name one control variable in this investigation. **(1)**

b How is the force being applied to the wire calculated? **(1)**

c Explain how the student should use the apparatus to determine if the wire obeys Hooke's law. **(5)**

d The student also wants to know if the sample behaves elastically. Explain how the student can improve the investigation and discover whether or not the wire behaves elastically. **(3)**

(Total for Question 3 = 12 marks)

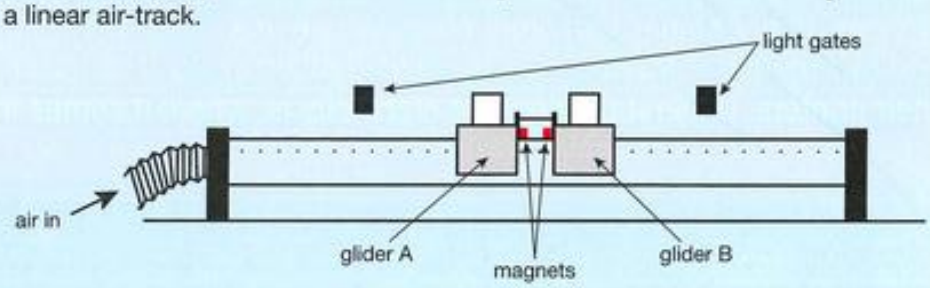
PHYSICS ONLY

SKILLS CRITICAL THINKING 7

4 This question is about momentum.

a State whether momentum is a scalar or a vector quantity and explain your answer. **(3)**

The following apparatus is used to investigate the momentum of two gliders on a linear air-track.



The gliders are fitted with magnets with like poles facing each other, so that they repel each other, and tied together with a cotton thread. This cotton thread will be burnt using a flame later in the investigation. 5 cm square cards are attached to both gliders – these will pass through the light gates. Digital timers time how long each card takes to pass through the light gates. The gliders are initially at rest.

SKILLS PROBLEM SOLVING

b State the initial momentum of the gliders. **(1)**

SKILLS INTERPRETATION 6

c i Draw a labelled diagram showing the forces acting on glider A before the thread is burnt. **(4)**

SKILLS CRITICAL THINKING 8

ii State Newton's third law and explain how it applies to the horizontal forces on the gliders when the thread is cut with the flame. **(3)**

SKILLS REASONING 6

iii Describe what happens to the two gliders when the thread is cut. **(1)**

SKILLS PROBLEM SOLVING

d After the thread is cut the card on glider A passes the light gate in 1.25 s.

6

i Calculate the speed of glider A as it passes through the light gate. **(2)**

10

ii Glider A has a mass of 500 g and glider B has a mass of 800 g. Calculate how fast glider B will be moving, explaining your method and any assumptions you make. **(5)**

SKILLS REASONING 8

e The experiment is repeated but the two gliders are tied closer together with a shorter cotton thread. Explain what difference this would make to the result of cutting the thread, if any. **(3)**

(Total for Question 4 = 22 marks)

HINT

Think about magnetic forces.

END OF PHYSICS ONLY